Generally barley is used in the production of malt. Other grains used in the preparation of malt includes wheat, jowar and ragi.

The process of malting of cereal grains consists of the following steps:

1. Selection of grain and cleaning.

2. Steeping in cold water for 36 hours with 2 to 3 changes in water.

3. Germination: The grains are spread on wire mesh trays and kept for 3 days. Water is sprinkled over each of these trays.

4. Kilning: The germinated grains are dried at slow rate on kilns.

Amylase rich food (ARF) is germinated cereal flours which are extremely rich in the enzyme alpha-amylase. ARF are excellent weaning foods because they reduce the bulk of weaning foods and are energy dense. Malt is used in commercial proprietary foods, breakfast cereals, malted milk confectionaries, infant foods, bakery products and in brewing.

2.6. Cereal cookery

2.6.1. Gelatinisation

Starch granules do not dissolve readily in cold water but they will form a temporary suspension with the starch tending to settle out as soon as the mixture is allowed to stand. When heated with water, the grains absorb water and swell. This process is called as gelatinisation.

2.6.2. Gluten formation

Although all cereals are more or less similar in protein content, the unique presence of glutelin and gliadin in the wheat makes it suitable for certain recipes. Glutenin or glutelin is the protein which gives toughness and rubberiness to gluten. Gliadin gives elasticity. Due to its elastic properties, the dough can be rolled to prepare chapathi or poori.



Fig 2.8: Gluten formation (Chapathi and Poori)

2.6.3. Dextrinisation

Dry heat also brings changes to starch granules through a process known as dextrinisation. If a starch product is subjected to dry heat, carbohydrate compounds called dextrins are formed. When these are dissolved in water they have a sweet taste. Colour and flavor changes also occur. Extensive dextrinisation reduces the thickening power of starches. Eg. Toasted bread.

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